Senior Thesis

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Abstract

Human land use, primarily through farming and its resulting land disturbances impacts the soil sink of atmospheric hydrogen maintained by soil microbial communities; this thesis' aim is to assess, both through measurements of soil H2 uptake rate and bacterial genomics, what governs this rate. Based upon preliminary agricultural data that I assess this summer, it appears that tilling has a significant impact on agricultural soils' uptake ; I would like to study this further by comparing farming practices and the resultant microbial behaviors in comparison to undisturbed, non-agriculture soils.

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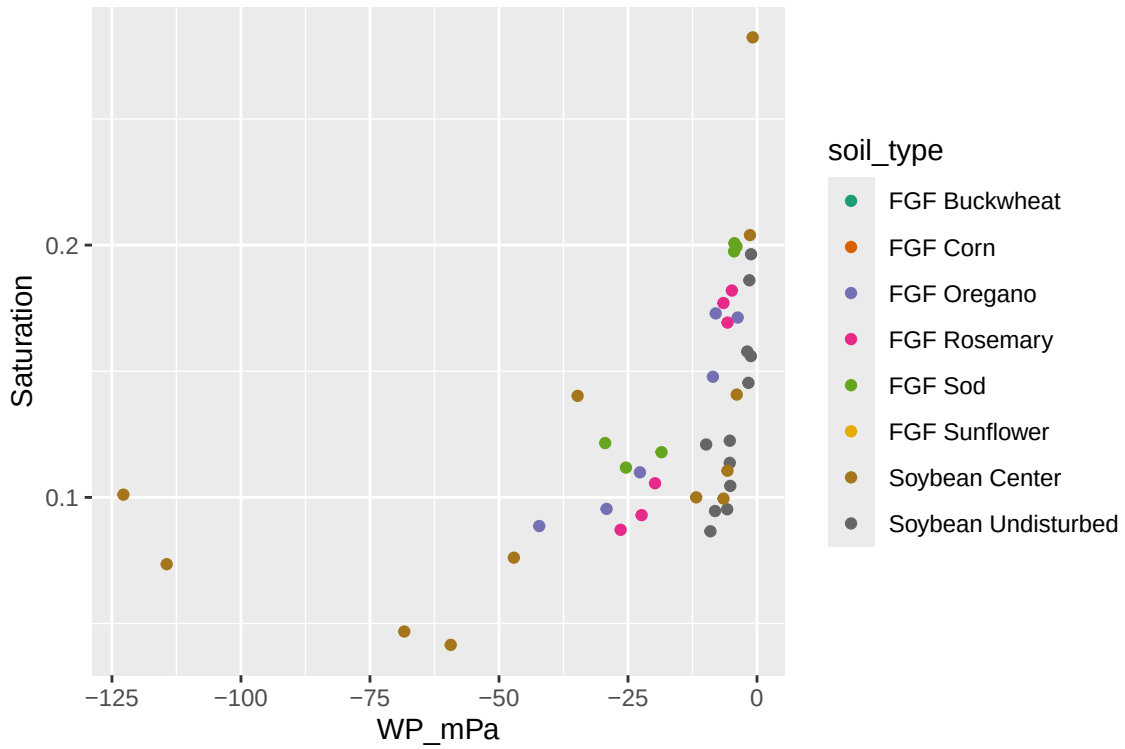
Introduction

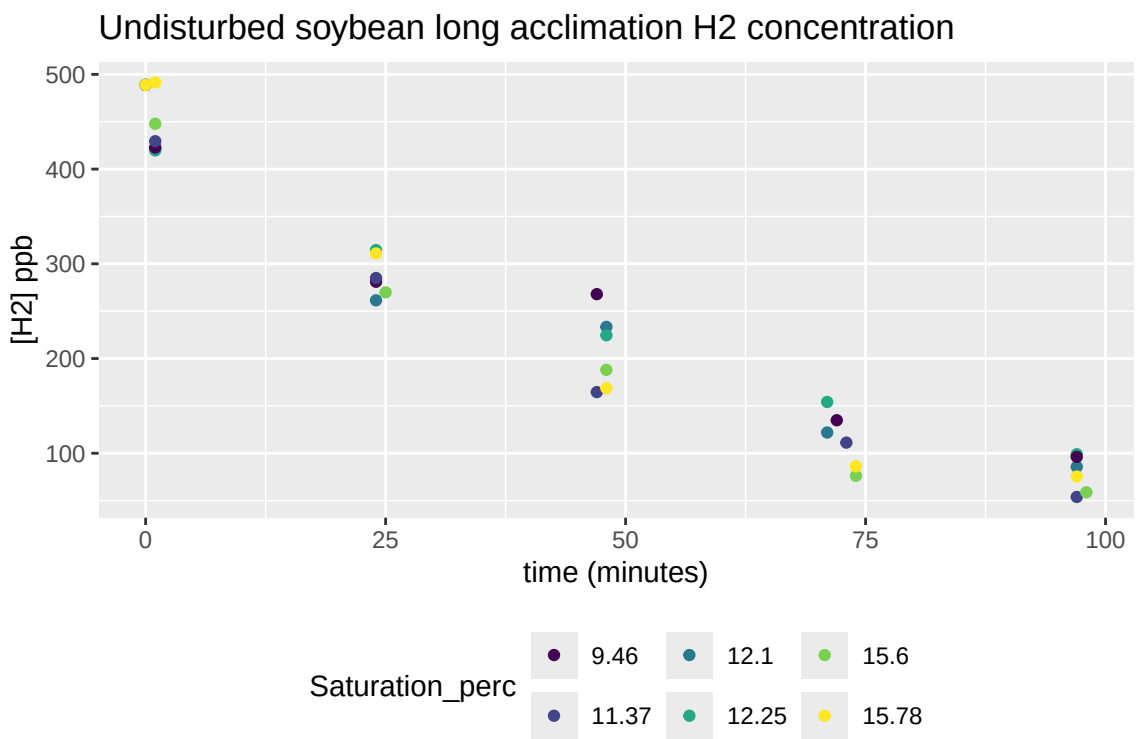
TBD Sand *et al.* (2023) says

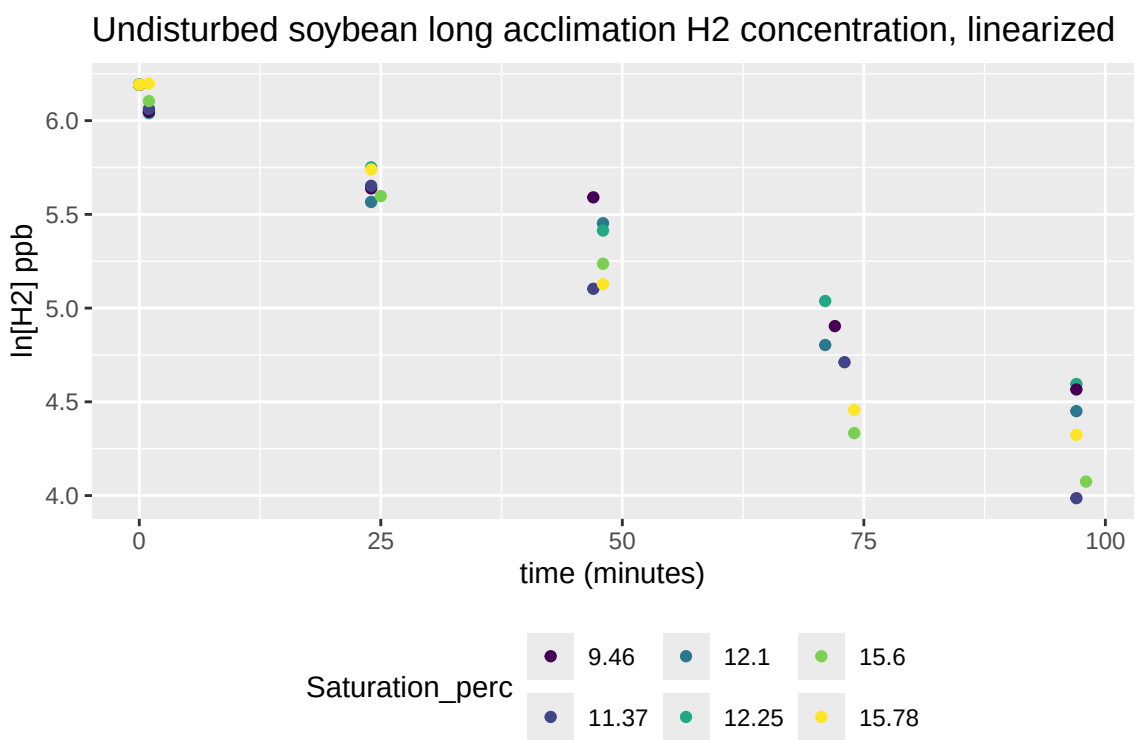
Methods

Soy - Undisturbed - Long

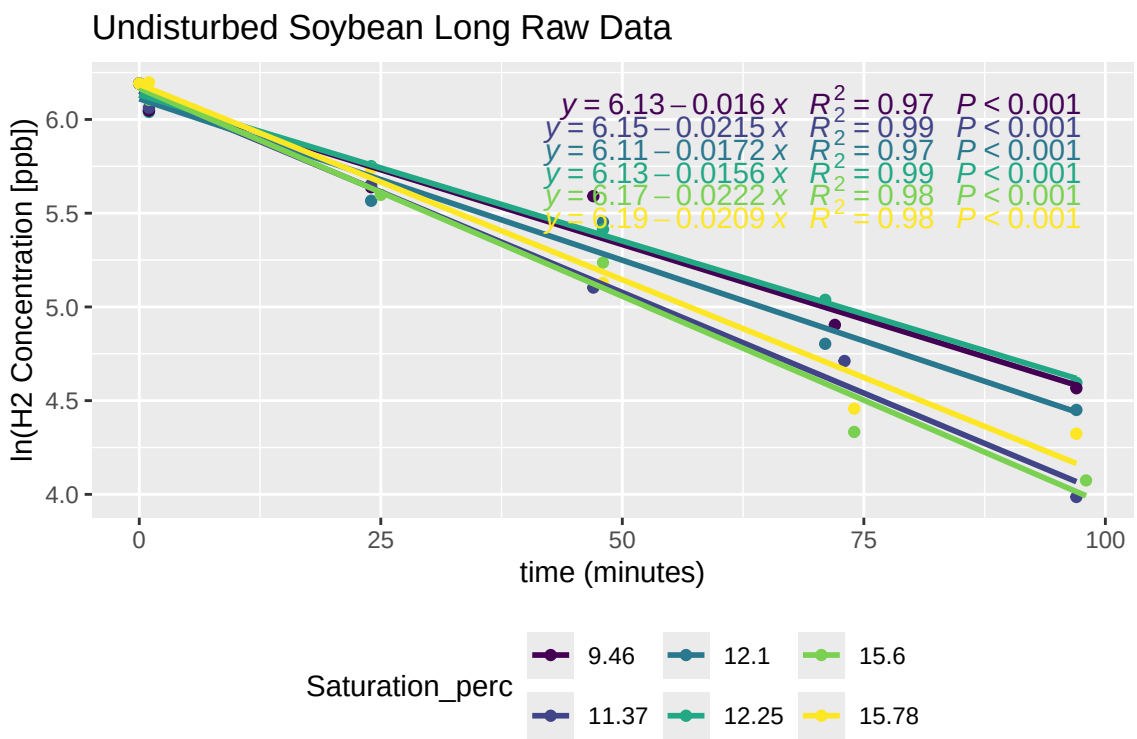
```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



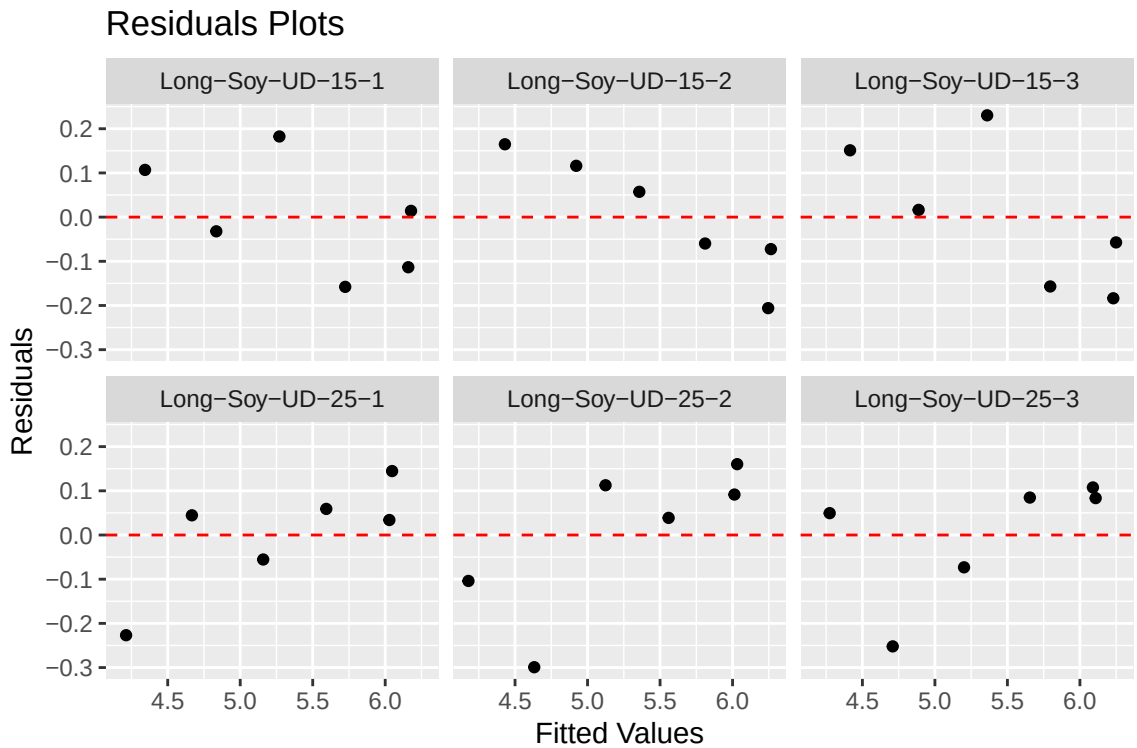





```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## Warning: `fortify(<lm>)` was deprecated in ggplot2 4.0.0.
## i Please use `broom::augment(<lm>)` instead.
## i The deprecated feature was likely used in the ggplot2 package.
##   Please report the issue at <https://github.com/tidyverse/ggplot2/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Of note: The fits are really good here, with very small residuals and R^2 values. The linearization seems to be effective without accounting for the asymptotic behavior near the tails, meaning that equilibrium is not sufficiently reached by the end of these incubation experiments within the time frame. I have included the fitting equations in the graph along with the R^2 values - the p values don't mean much because they only compare these lines to a slope of 0, which is obviously out of the question.

```
## Adding missing grouping variables: `bottle_id`
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
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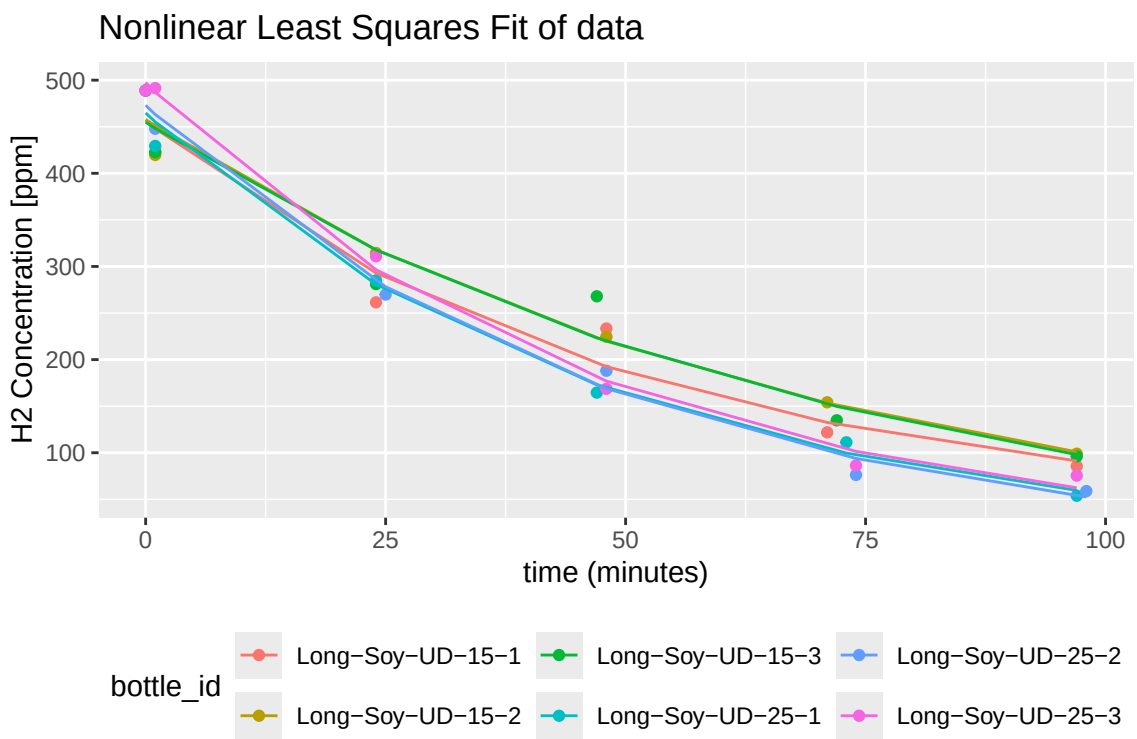
```

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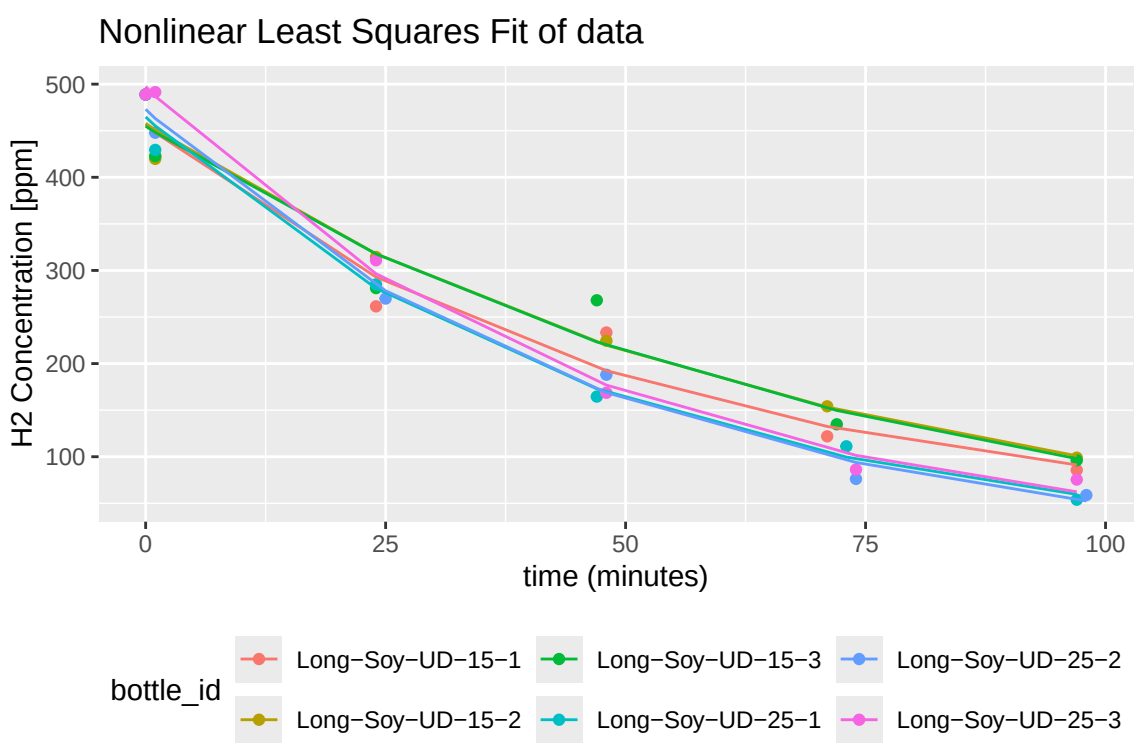
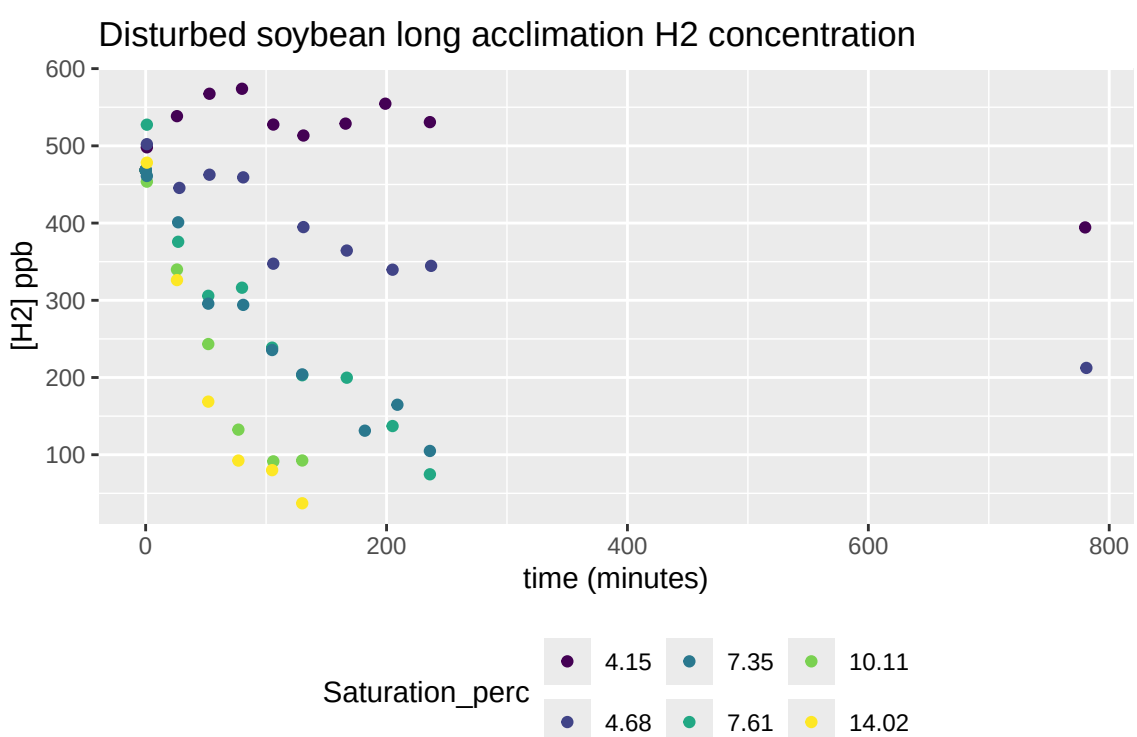


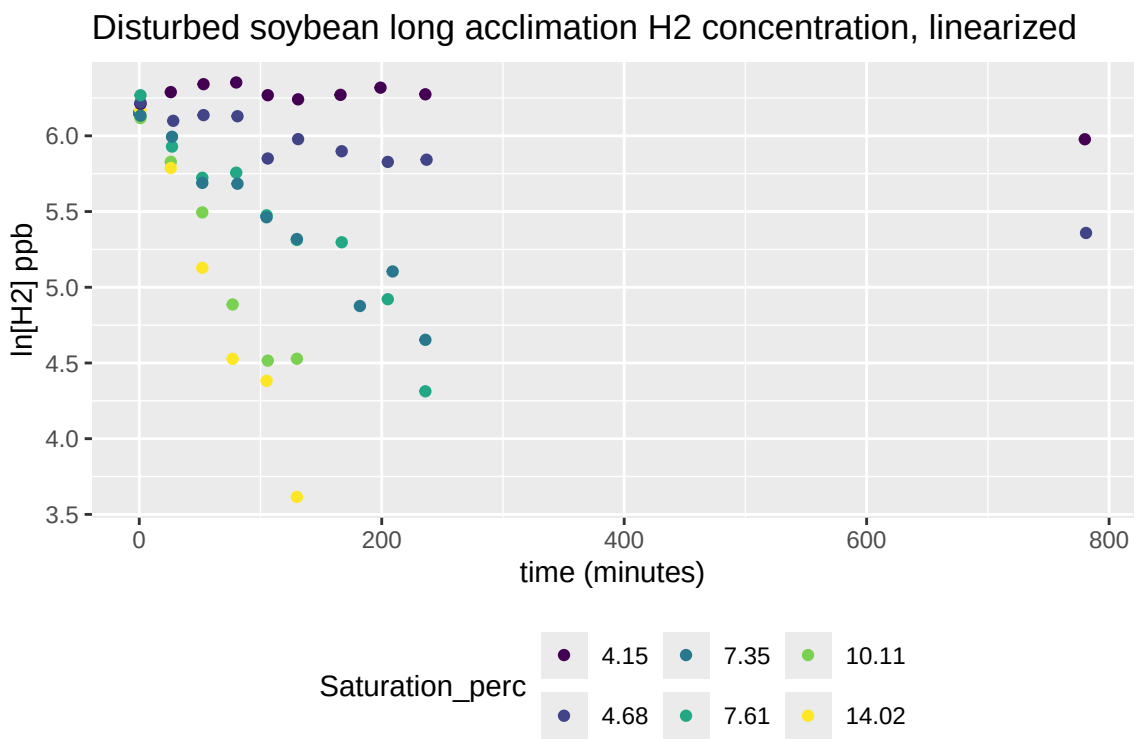
This nonlinear fit is less interpretable than the linearized fits seen above, because, as the

data tables show, it is often guessing that the asymptote line is below 0 ppb of H2, which is impossible. The earlier linearized fitting methods make more physical sense and don't seem to diverge from linearity based upon the residuals plot, but they *may* seem to diverge from homoscedasticity - the residuals plots for each don't seem totally amorphous, but they may be a result of the very small sample sizes per bottle. Regardless, heteroscedasticity does not introduce bias into the model (it doesn't sway the regression coefficient one way or another), but it does mean our models are less precise than the high R^2 values would make them appear to be.

There are methods that I want to use to seek to address this - the first being a new-to-me method of regression called a robust regression, another being perhaps a background subtraction of the hydrogen background in the nitrogen carrier gas. Most importantly, I think more trial runs will show us how they are behaving at a grander scale and if there are systemic deviations from the assumed linearity of our natural log linearization.

Soy - Disturbed Center - Long






```
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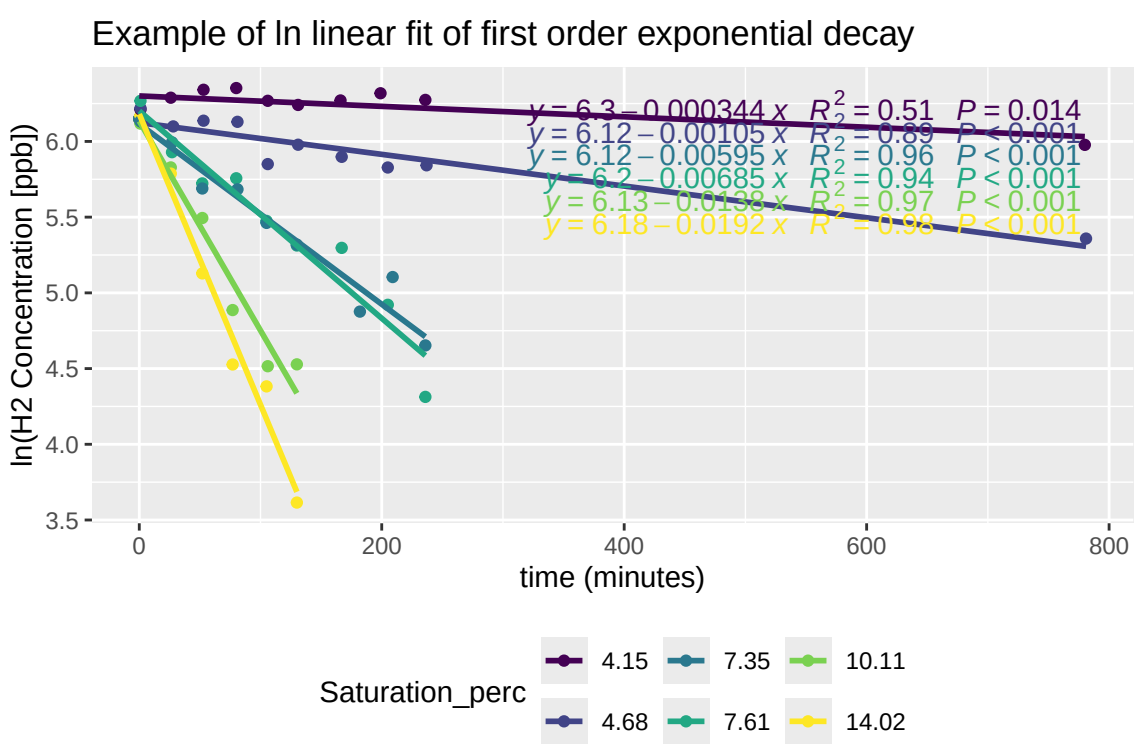

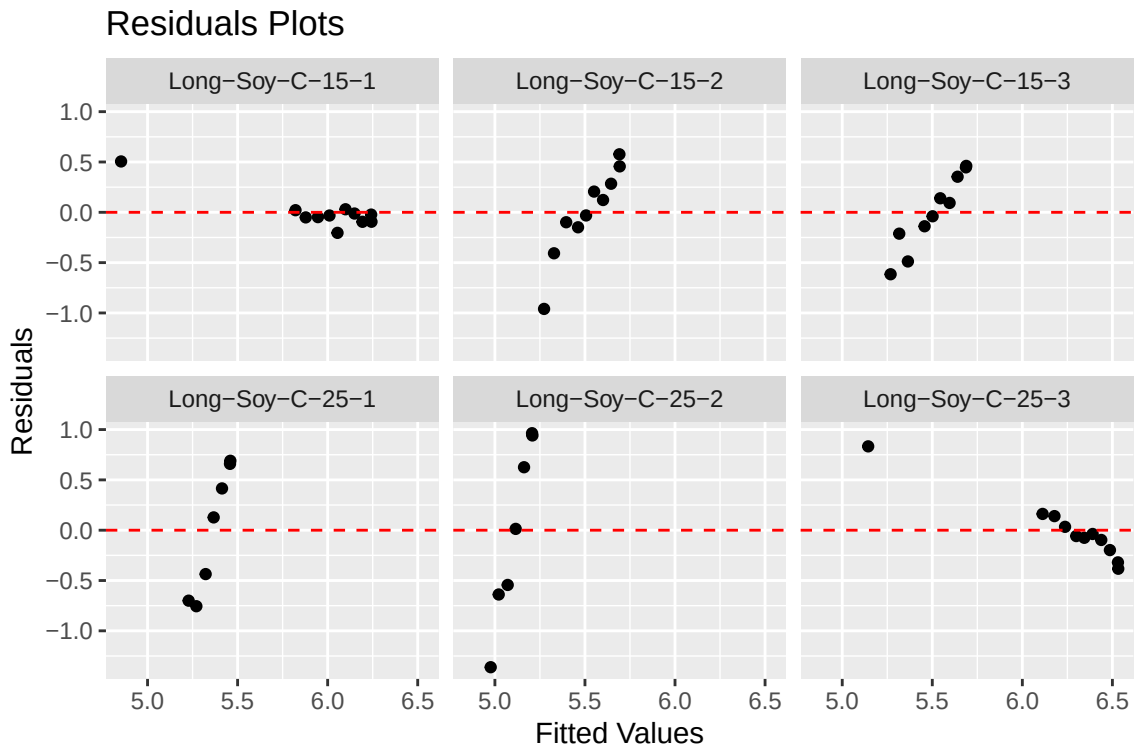


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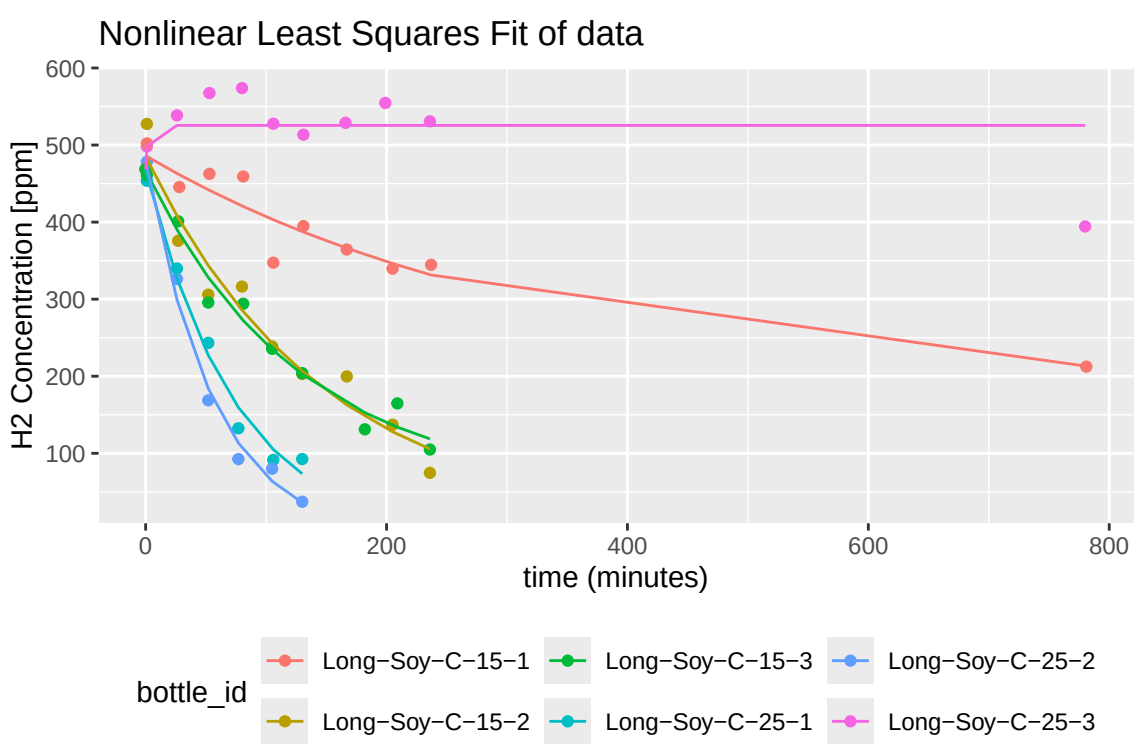


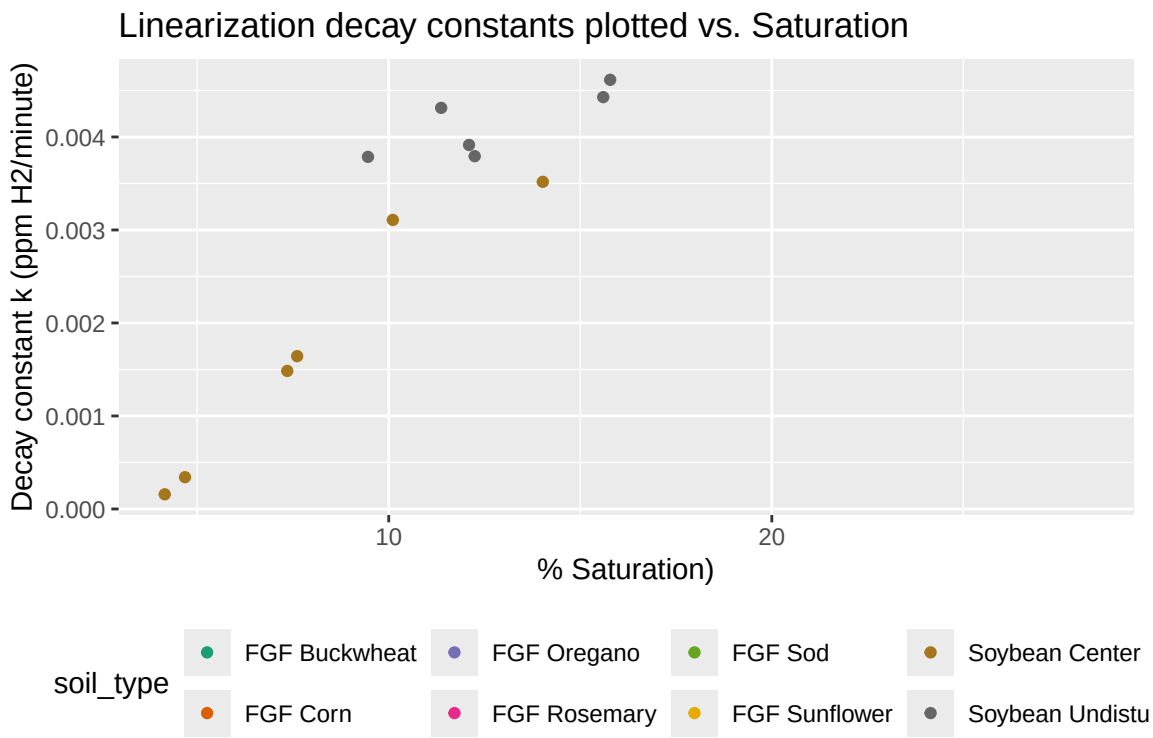
Good Soy Runs

Fairgrown Farms

```
## [1] 77
```

Kinetic Rate Data Summary

```
## Warning: Removed 54 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

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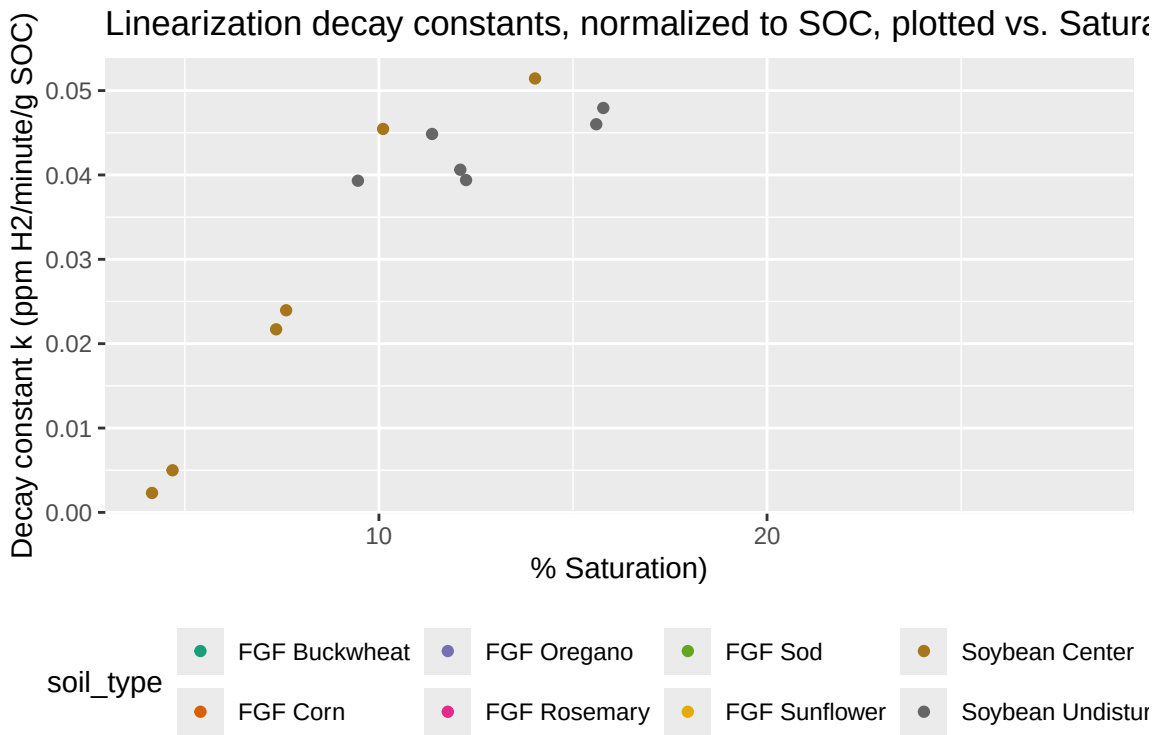


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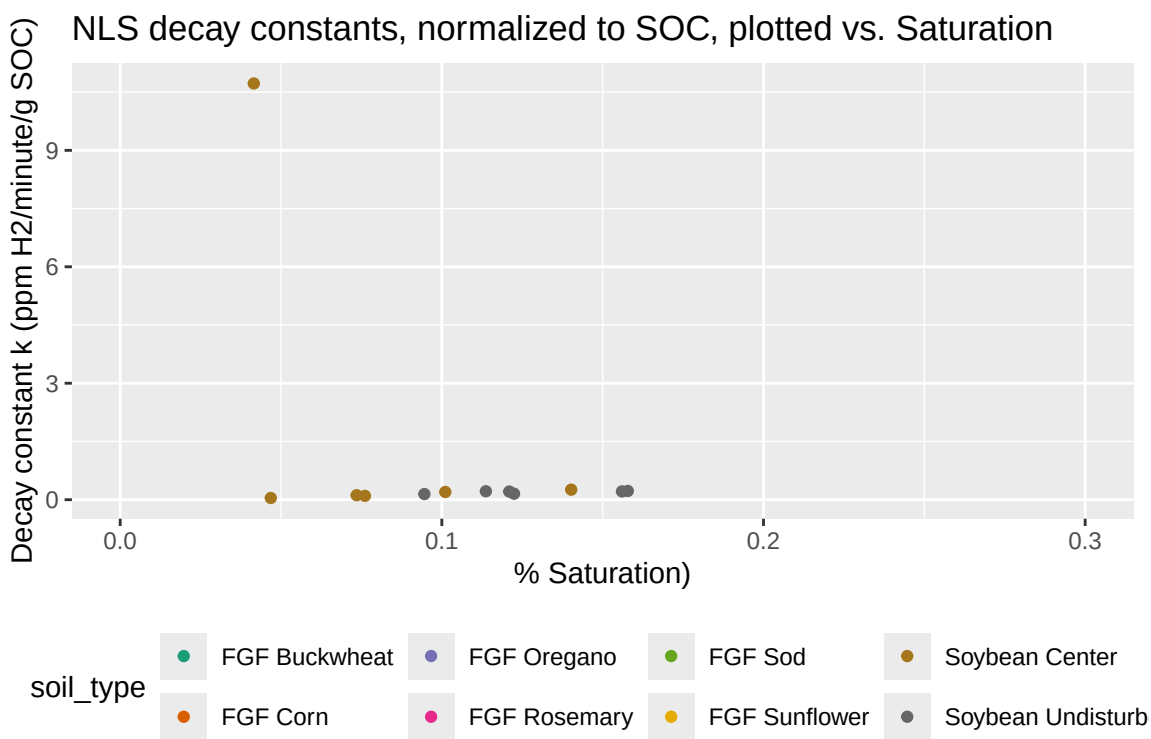

Results

Include draft work from CEE523 on equation writing.

Conclusion

References

Appendix

Sand, M., Skeie, R.B., Sandstad, M., Krishnan, S., Myhre, G., Bryant, H., Derwent, R., *et al.*, 2023. A multi-model assessment of the Global Warming Potential of hydrogen. *Communications Earth & Environment*, **4**, 203. doi:10.1038/s43247-023-00857-8